DRONA: Lightweight Mutual Authentication Protocol for Drones Using PUF

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***Abstract*—Drones commonly known as Unmanned Aerial Vehicles (UAVs) are becoming very popular nowadays for their wide applicability in areas ranging from delivery systems to aerial photography to automated military applications. Nevertheless, communication security between a UAV and its base station is critical for successfully completing its task without leaking sensitive information either to the adversaries or to unauthenticated users. UAV systems are especially vulnerable to physical and node tampering attacks. Further, since UAV devices are generally equipped with small batteries and limited memory storage, lightweight security techniques are best suited for them. Addressing these issues, a lightweight mutual authentication scheme based on Physical Unclonable Function (PUF) for UAV networks is presented in this paper. We present a formal security analysis to show that our protocol provides various security features and is resilient against many security attacks. We also show a performance comparison of our protocol with the state-of-the-art protocols for securing UAV networks based on computation time and memory usage.**

***Index Terms*—UAVs, physical security, mutual authentication, security protocol, privacy, PUFs.**

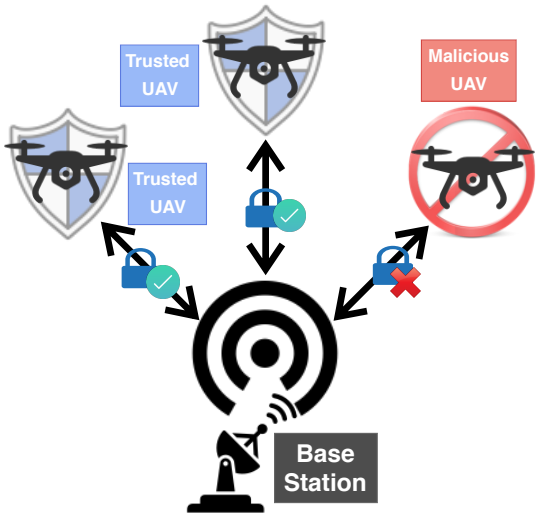

Fig. 1: System model

I. INTRODUCTION

Around the world, Unmanned Aerial Vehicle (UAV) technology is being developed and deployed at a very rapid pace. From originally being used in military applications, these tools are being adopted across the government and corporate entities for providing a wide range of services including traffic management, goods delivery, inventory tracking etc. With many governments easing regulations for UAV usage, the interest in this emerging technology is growing leaps and bounds leading to newer fields of applications being developed. UAV based networks now can be used for traffic surveillance in the next generation Intelligent Transportation Systems [1], [2].

Although the UAVs provide various benefits, due to being deployed in the open environment they are also prone to several security threats. Primarily, the threats to UAVs come in two types- natural phenomena, and those caused by human interference [3]. The quality of data transmitted is influenced by the natural phenomena, whereas human interference affects the security aspect of the transmitted data. Some of these security threats could be modifying the communication data,device capture, eavesdropping attack, flying away with the device etc. Inter-UAV communication and that of UAVs with the base station being wireless in nature are thus prone to several of the above malicious attacks.

Node authentication is the first security aspect of any IoT network, which should be met before beginning a secure communication session in the network. The existing Internet security techniques implement authentication using stored cryptography keys. UAVs being deployed in open air are prone to device capture, and thus to secret keys being exposed. Further, IoT devices generally have limited memory and computation capability, thus storing the secret keys and executing standard cryptography algorithms is also a challenge. This paper addresses the problems associated with generation and storage of secret keys for authenticating devices in the UAV network. Addressing computation capability, a light-weight mutual authentication protocol is presented here, well suited to the UAV networks.

With the rapid development of integrated circuit technology in the recent years, PUFs have been found to be very promising in a wide range of security applications [4–6]. They utilize the inherent randomness introduced during the manufacturing process of a silicon device, thus making it very difficult to clone or reproduce them [7–9]. Two different PUFs manufactured using the same fabrication process and with the same configuration are expected to produce different responses given the same

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challenge. The above property of the PUFs is exploited here for generating session keys in our protocol.

The organization of the rest of the paper is as follows. Section II discusses the related works in securing UAV networks. The system model, security goals, attack model and assumptions for our system model are discussed in Section III. In Section IV, we present our mutual authentication protocol (DRONA) based on PUFs. In Section V, we subject our protocol to formal security analysis. Performance analysis of our protocol and comparison with other state-of-the-art security protocols for UAV networks is shown in Section VI. We finally conclude the paper in Section VII.

II. Related Works

UAV Networks (UAVNs) differ greatly from the other traditional wireless networks like Mobile Ad-Hoc Networks (MANETs) due to greater node mobility and thus the existing security solutions for MANETs and other traditional wireless networks cannot be adapted to UAVNs. He et al. have explained the essential features of UAV communications and identified the requirements for UAV security protocols in [10]. They have shown that traditional security techniques such as anomaly detection induce significant delays and therefore cannot be used for time-critical applications like those in UAVNs. Thus, they establish the need for UAV security protocols which take into account the trade-off between network performance and security strength.

Hooper et al. have presented a multi-layer framework in [11] for defending commercial UAV systems from Address Resolution Protocol (ARP) cache poisoning, buffer overflow and Denial of Service (DoS) attacks. Blazy et al. have considered a strong adversarial model which can perform fault injection, side-channel and physical attacks on a UAV [12]. In their scheme, they store a single initial key which is used to generate subsequent keys using a keyed hash function. They propose to use multiple streams of such key sequences to prevent an attack. But, if a powerful attacker gains just one key of each stream, he/she could very easily keep track of all the key streams simultaneously thereby rendering all further communication insecure. Chen et al. have proposed the Direct Anonymous Attestation with Mutual Authentication (MA-DAA) scheme in [13] for use in network-connected UAV systems. Compared with the existing Direct Anonymous Attestation (DAA) schemes, MA-DAA scheme is well suited to the low bandwidth and computation capabilities of the UAV networks. However their solution is based on Trusted Platform Modules (TPMs), which are specialized expensive security co-processors that need to be integrated into the systems leading to higher costs.

Much of the recent research to secure UAV communications is focused on physical layer security (PLS) [14–17]. PLS exploits various wireless channel characteristics and well-defined transmission schemes, and degrades the communication channel of an eavesdropper compared to a legitimate user [18]. Some amount of secrecy is always lost to an eavesdropper in any PLS scheme, thus making it unsuitable for highly sensitive applications such as military.

PUFs have been shown to be very effective in securing wireless and unmanned IoT environments recently [19, 20]. Chen and Willems propose a secret key generation protocol based on PUF in [21]. In another work [22], researchers have shown the feasibility of effectively extracting the security keys using light weight fuzzy extractors well suited to the IoT environment. The proposed protocol in this paper uses PUF for generating secret keys, which ensures privacy. The major contributions of this paper are: (i) Lightweight PUF based mutual authentication protocol for energy and memory constrained UAV networks, (ii) Generation of unique session keys for each authentication session and (iii) Security and privacy of session keys generated using PUFs.

III. System & Threat Model

A. System Model

The system model used in this paper consists of three entities namely legitimate UAVs, eavesdropper UAVs, and a base station (BS) as shown in Fig. 1. UAVs have limited memory and computational capability as compared to the BS. In this model, multiple UAVs may be connected to a single BS. The computational device on every UAV is equipped with a PUF, which is used for generating a 32-bit response for a 32-bit challenge input to it. The UAV itself does not store any secret keys, rather performs a key expansion on the generated response to help generate session keys. Both the UAV and the BS are capable of performing this mutually agreed-upon key expansion procedure which is based on a public non-linear function F. Whenever a new UAV wants to register itself with the BS, it's 32-bit challenge response pair (C, R) is securely sent and stored in the BS. The BS is the only trusted authority in this network, and therefore (C, R) pairs for all UAVs are stored only in the BS. Further in this model, for each UAV the BS starts with a single initial (C, R) pair which it acquires at the time of UAV registration. After this initial exchange, the UAV can function independently without needing any technical personnel or any secure channel. While the UAV generates (C, R) pairs on the fly upon each authentication attempt, the new (C, R) pair is securely sent and stored in the BS. The ID of the authenticated UAV is dynamically changed at the end of authentication both in the BS and in the UAV, and this new ID is used for subsequent communication until it is changed at the end of next authentication. Nothing else is assumed in further communication.

B. Security Goals

Keeping the following security goals in view, this protocol has been proposed.

1) Achieving mutual authentication between the individual UAV nodes and the BS.
2) The communication between the UAV and the BS must be confidential.
3) The protocol must be capable of generating a unique session key for each authentication session.
4) The session keys generated must be safe against physical attacks like UAV node capture and node tampering.

TABLE I: Notations

Notation	Description
U_i	ID of ith UAV
B_i	ID of ith BS
$\|$	Concatenation
$\oplus$	XOR operation
$MAC()$	Message Authentication Code
N_A, N_B, N_C	Nonces generated
Q	Cipher text prepared by BS
$F()$	Any public non-linear function with 32-bit input and output
$H()$	Hash function

5) If the communicated messages are tampered, the receiver, either BS or UAV, must be able to detect the tampering, and infer the presence of an attacker.
6) No unauthorized entity must be capable of tracking the ID of a UAV.

C. Attack Model

We assume that an adversary, which can be either an authenticated UAV or an intruder UAV can eavesdrop on any communication between the legitimate UAV and the BS. An attacker may try to tamper with the data being communicated in the UAV network, impersonate as a valid UAV or BS, store and replay the messages from previous sessions, inject its own messages in the communication link, or initiate new sessions and authenticate with the BS. Moreover, in this model, the adversary may also have the capability to physically capture a flying UAV. This paper presents a protocol to successfully eliminate these threats in a lightweight and efficient manner by using PUFs.

D. Assumptions

We discuss here a few assumptions that are made in this paper. Each UAV node in the network is endowed with a unique PUF to support creation of session keys on the fly. UAV nodes have limited computational power and are memory constrained while the BS has no such constraints. In case of a UAV capture, any attempt to tamper with the PUF will render the PUF unusable and therefore the captured UAV will not be able to authenticate with the BS. Table I lists the notations used in this paper and their descriptions.

IV. Proposed PUF-based DRONA Protocol

This section presents the proposed DRONA protocol between a UAV and the BS which is depicted in Fig. 2. When a UAV U_1 wants to authenticate with a nearby BS B_1, it sends its ID U_1 and a nonce N_A to B_1. B_1 checks within its memory if U_1 exists and whether nonce N_A is fresh, i.e, it does not match the N_A generated in the previous authentication. If either of the condition fails, the authentication request by U_1 is terminated. Using U_1, the base station B_1 finds the corresponding 32-bit challenge-response pair (C, R) from its memory. Using a known public key expansion procedure [23], it generates m 32-bit keys called $K_1, K_2, K_3, ..., K_m$ from the response R. It then generates a 32-bit nonce N_B, which along with N_A is the information to be encrypted, as $N_A||N_B$. The function F could be any publicly available non-linear function which takes a 32-bit input and gives a 32-bit output, and it must be mutually agreed upon by both the BS and the UAV. The value of m could be any even number greater than 2, which can be decided upon based on the speed of the algorithm required.

$$
\begin{aligned}
X_1 &= N_A \oplus K_m \\
Y_2 &= N_B \oplus F(X_1) \oplus K_{m-1} \\
X_3 &= X_1 \oplus F(Y_2) \oplus K_{m-2} \\
Y_4 &= Y_2 \oplus F(X_3) \oplus K_{m-3} \\
&\cdots \\
Y_{m-2} &= Y_{m-4} \oplus F(X_{m-3}) \oplus K_3 \\
X_{m-1} &= X_{m-3} \oplus F(Y_{m-2}) \oplus K_2 \\
Y_m &= Y_{m-2} \oplus K_1
\end{aligned}
$$

The final cipher-text Q to be sent to U_1 and the value m is encoded in the following way.

$$
\begin{aligned}
Q &= (Y_m \| X_{m-1}) \oplus (K_m \| K_{m-1}) \\
N &= m \oplus K_1
\end{aligned}
$$

B_1 then sends C, Q, N and a Message Authentication Code $MAC(B_1 \| Q \| m \| N_A \| N_B)$ to U_1. Within the MAC, the first parameter verifies the identity of the BS, while data integrity is ensured by the second and third parameters. Freshness of the source (B_1 in this case) is identified by the last two parameter N_A and N_B. On receiving the message from the BS, U_1 generates the response R by passing the received challenge C through the PUF as given in (1). Using the same key generation and expansion procedure as used by the BS, it regenerates m 32-bit keys $K_1, K_2, K_3, ..., K_m$ from the response R. It calculates m and $Y_m \| X_{m-1}$ as shown below.

$$R = PUF(C) \tag{1}$$

$$m = N \oplus K_1 \tag{2}$$

$$(Y_m \| X_{m-1}) = (K_m \| K_{m-1}) \oplus Q \tag{3}$$

This is divided into two 32-bit strings to get Y_m and X_{m-1}. The decryption procedure is identical to the encryption procedure as can be seen in the following set of equations.

$$
\begin{aligned}
L_1 &= Y_m \oplus K_m \\
T_2 &= X_{m-1} \oplus F(L_1) \oplus K_{m-1} \\
L_3 &= L_1 \oplus F(T_2) \oplus K_{m-2} \\
T_4 &= T_2 \oplus F(L_3) \oplus K_{m-3} \\
&\cdots \\
T_{m-2} &= T_{m-4} \oplus F(L_{m-3}) \oplus K_3 \\
L_{m-1} &= L_{m-3} \oplus F(T_{m-2}) \oplus K_2 \\
T_m &= T_{m-2} \oplus K_1
\end{aligned}
$$

Now, the original information $N_A||N_B$ is recovered as $T_m \| L_{m-1}$. From this, N_B is retrieved and the MAC is recomputed. Thus, the source of the message, its freshness

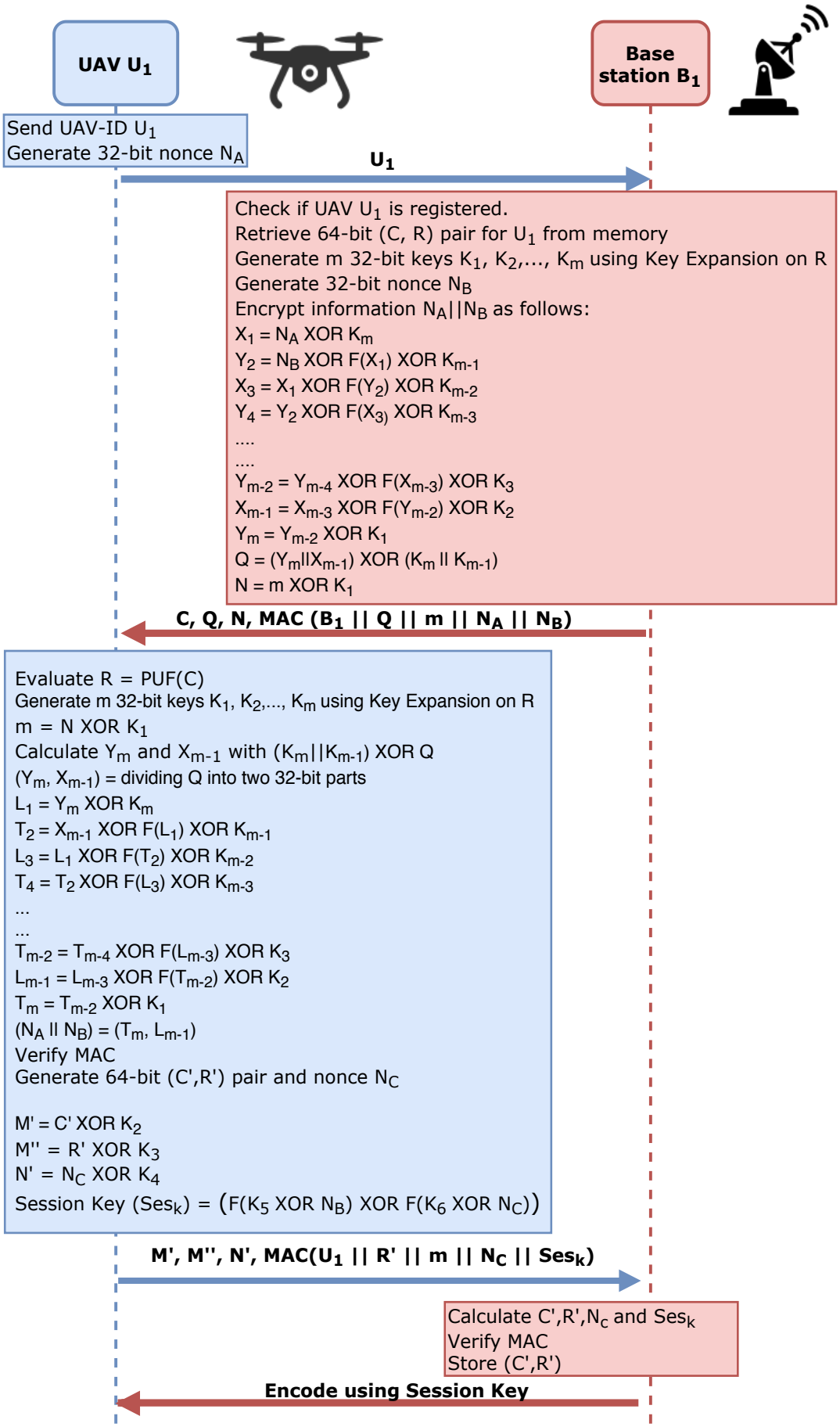


Fig. 2: DRONA protocol between a UAV and the Base Station

and integrity are verified. In case MAC verification does not succeed, B_1 terminates the authentication of U_1. If verified, a new 32-bit challenge-response pair (C', R') is generated by U_1 using its PUF. It also generates a new Nonce N_C. This newly generated information is encoded into $M', M''andN'$ as follows.

$$M' = C' \oplus K_2$$
$$M'' = R' \oplus K_3$$
$$N' = N'_C \oplus K_4$$

The session key Ses_k with which further communication will take place is computed as shown below.

$$Ses_k = ((F(K_5 \oplus N_B) \oplus F(K_6 \oplus N_C)) \tag{4}$$

M', M'', N' and a new MAC with five parameters is sent to B_1. B_1 calculates C', R', N_C and Ses_k as shown in the following equations.

$$C' = M' \oplus K_2$$
$$R' = M'' \oplus K_3$$
$$N'_C = N' \oplus K_4$$
$$Ses_k = ((F(K_5 \oplus N_B) \oplus F(K_6 \oplus N_C))$$

Now, B_1 has all the parameters to verify MAC. If the verification fails, authentication is terminated by B_1. If verified, it stores the new 32-bit challenge response pair (C', R') for UAV U_1 in its memory replacing the previous entry (C, R). Until U_1 moves out of B_1's range, their communication is encrypted using Ses_k. Thus, mutual authentication has been achieved. Both U_1 and B_1 compute the new Alias ID (AID) for U_1 and update it in their storage.

$$AID = H(U_1||K_7) \tag{5}$$

Changing the ID of the UAV using a suitable hash function as shown in equation (5) ensures untraceability. An adversary will not be able to match the AID with any previous IDs.

V. Security Analysis

A formal verification of the protocol guarantees that an adversary cannot gain or modify any information communicated in the network. We have used Mao Boyd logic [24] for verification of our protocol's security. In our proof we invoke the ***authentication***, ***nonce-verification***, ***confidentiality***, ***super-principal***, ***intuitive*** **and** ***good-key inference rules*** of Mao Boyd logic. Throughout the verification process, we denote the UAV U_1 by U and the Base station B_1 by B.

We first prove the statement "U believes N_B is a good secret between U and B". The following proof is shown pictorially in Fig. 3a. The (C, R) pair of U is stored in B, therefore it can be said that U believes R is a good secret between U and B. This is written in Mao Boyd logic as shown in (i). Using R, the UAV U is able to decipher Q in message 2 of the protocol to get $N_A \parallel N_B$. Since U is aware that N_A is contained in the first 32 bits, it can extract N_A. Therefore, we have the statement "U sees N_A with decipher key R" which is written symbolically in (ii).

$$U \models U \overset{R}{\leftrightarrow} B \tag{i}$$
$$U \overset{R}{\triangleleft} N_A \tag{ii}$$

Applying the ***authentication rule*** in [24] to statements (i) and (ii), we arrive at the statement "U believes B encrypted N_A using R" which is statement (iii). Since U generates a new nonce N_A each time, we have the statement "U believes N_A is fresh" which is (iv). Next, we apply the ***nonce-verification rule*** to statements (iii) and (iv) to get (v) which is "U believes that B believes that R is a good secret between U and B".

$$U \models B \overset{R}{\mid\!\sim} N_A \tag{iii}$$
$$U \models \#(N_A) \tag{iv}$$
$$U \models B \models U \overset{R}{\leftrightarrow} B \tag{v}$$

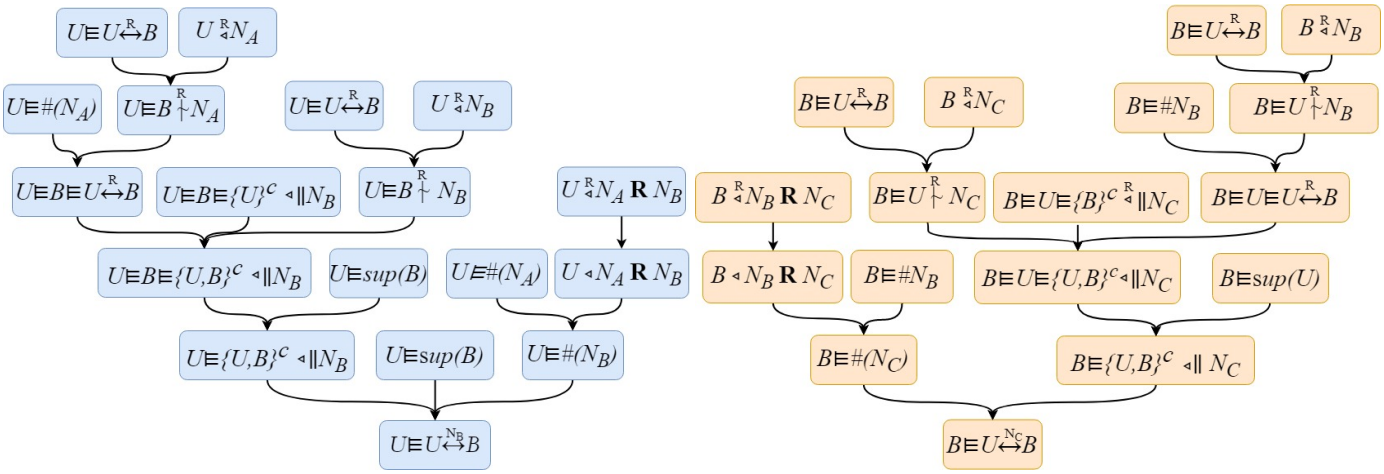
Using R, the UAV U decrypts Q and acquires N_B in message 2 of the protocol. Therefore, we next have the statement "U sees N_B with decipher key R" which is (vi). Applying the ***authentication rule*** to (i) and (vi), we get (vii) which is "U believes B encrypted N_B using R". Since B is known to generate a new nonce N_B each time, U is aware of the fact that no one other than B could have seen N_B. Thus it can be said that "U believes that B believes that no one other than U has access to N_B" which is (viii). Applying the ***confidentiality rule*** to (v), (vii) and (viii) we get (ix) which states "U believes that B believes that no one other than U and B has access to N_B".

$$U \overset{R}{\triangleleft} N_B \tag{vi}$$
$$U \models B \overset{R}{\mid\!\sim} N_B \tag{vii}$$
$$U \models B \models \{U\}^c{\triangleleft} \parallel N_B \tag{viii}$$
$$U \models B \models \{U, B\}^c{\triangleleft} \parallel N_B \tag{ix}$$

It is assumed in the protocol that U believes that the base station B is secure and trusted or in other words we have the inference "U believes that B is the super-principal". which is (x). Applying the ***super-principal rule*** to statements (ix) and (x), we get (xi) which states "U believes that no one other than U and B has access to N_B".

To proceed further we need to understand a few definitions and rules of message idealization from [24] which are as follows:

- If a message does not have any symbols, it is an *atomic message (AM)*.
- An *AM* which is sent in one line and received in another by the origin node is called a *challenge*.
- If a *challenge* is present in a message sent to the origin node, it is a *replied challenge (RC)*.
- A *response* is an *AM* and an *RC* sent together by the node which sends the *response*.
- If an *AM* is neither a *challenge* nor a *response*, it is *nonsense*. All *nonsense* is discarded.



(a) U believes that N_B is a good secret known only to U and B

(b) B believes that N_C is a good secret known only to U and B

Fig. 3: Formal Security Proof

- If an *AM* behaves as a *challenge* as well as a *response* in a line, it is treated as a *response*.
- "*Response* **R** *RC*" is the combination of a *replied challenge* and its *response*.

In the first message of the protocol N_A is sent to B. In response to this, sends N_B in message 2 which is encrypted in Q. On decryption of Q, U obtains $N_A \| N_B$. Therefore, according to the above mentioned definitions and rules, N_A is a challenge and N_B is the response. Note that these are not the same challenge-response pair (C, R) of the PUF. Thus we have the statement "U can see the replied challenge N_A and the response N_B with decipher key R" which is (xii). Applying the ***intuitive rule*** to (xii), we get (xiii) which is "U can see the replied challenge N_A and the response N_B".

$$U \models sup(B) \tag{x}$$

$$U \models \{U, B\}^c \triangleleft \| \; N_B \tag{xi}$$

$$U \overset{R}{\triangleleft} N_A \; \mathbf{R} \; N_B \tag{xii}$$

$$U \triangleleft N_A \; \mathbf{R} \; N_B \tag{xiii}$$

On applying the ***fresh rule*** to (iv) and (xiii), we get the statement (xiv) which is "U believes N_B is fresh".

$$U \models \#(N_B) \tag{xiv}$$

$$U \models U \overset{N_B}{\leftrightarrow} B \tag{xv}$$

Applying the ***good-key rule*** to statements (x), (xi) and (xiv) we have proved the statement "U believes that N_B is a good secret known only to U and B". Similarly we can prove the statement "B believes that N_B is a good secret known only to U and B", as is shown in Fig. 3b. The statements "B believes that N_C is a good secret known only to U and B" and "U believes that N_C is a good secret known only to U and B" are shown in Fig. 4 and Fig. 5 respectively. In these figures, the logical AND operation between two statements is represented by a '$\bigwedge$'. Thus, we have shown that neither N_B nor N_C can be accessed by an adversary. Similarly, it can be proved that an adversary cannot gain access to R'.

$$\dfrac{\dfrac{B \models U \overset{R}{\leftrightarrow} B \bigwedge B \models U^c \triangleleft \| N_B \bigwedge B \overset{R}{\mid\!\sim} N_B}{B \models \{U,B\}^c \triangleleft \| N_B} \bigwedge B \models \#(N_B)}{B \models U \overset{N_B}{\leftrightarrow} B}$$

Fig. 4: B believes that N_B is a good secret known only to U and B

$$\dfrac{\dfrac{U \models U \overset{R}{\leftrightarrow} B \bigwedge U \models B^c \triangleleft \| N_C \bigwedge U \overset{R}{\mid\!\sim} N_C}{U \models \{U,B\}^c \triangleleft \| N_C} \bigwedge U \models \#(N_C)}{U \models U \overset{N_C}{\leftrightarrow} B}$$

Fig. 5: U believes that N_C is a good secret known only to U and B

If an adversary cannot obtain N_B, N_C and R', it cannot make sense of the communicated data. The Mao Boyd formal proof discussed above has proven the secrecy of N_B, N_C and R' which is regardless of the kind of attack used by the adversary such as masquerade attack, man-in-the-middle (MITM) attack, and replay attack. Note that even if an attacker captures the UAV, the property of PUF mentioned in section I ensures that it cannot gain any legitimate (C, R) pair. Additionally, there are no secrets stored on the UAV itself. This guarantees physical security and protection against node tampering attack. The use of an Alias ID (AID) as in equation (5) guarantees untraceability. Also, a unique session key is generated in each session of authentication.

VI. Comparison and Analysis

A. Security Comparison

This section provides a comparative analysis of our protocol with other existing security schemes available for UAV networks [25–30]. The comparison is based on five important

TABLE II: Comparison of security features and protection against attacks

	Security Features					Protection Against Attacks			
Protocols	**SF1**	**SF2**	**SF3**	**SF4**	**SF5**	**PA1**	**PA2**	**PA3**	**PA4**
[25]	✗	✗	✗	✗	$*$	✗	✗	✓	✗
[26]	✓	✗	✗	✗	$*$	✓	✗	✓	✗
[27]	✓	✗	✗	✓	✗	✓	✗	✗	✓
[28]	✓	✗	✓	✗	✗	✗	✗	✓	✗
[29]	✓	✗	✓	✗	$*$	✓	✗	✓	✗
[30]	✓	✓	✓	✗	$*$	✓	✓	✓	✗
Ours	✓	✓	✓	✓	✓	✓	✓	✓	✓

SF1- Mutual Authentication, SF2- Untraceability, SF3- Multiple Session Keys
SF4- Physical Security, SF5- Security Proof ($*$ Informal Analysis)
PA1- Masquerade Attack, PA2- MITM Attack
PA3- Replay Attack, PA4- Node Tampering Attack

security features labelled in the table as SF1, SF2, SF3, SF4 and SF5 which refer to mutual authentication, untraceability, multiple session keys, physical security and security proof respectively, and based on protection against common security attacks labelled as PA1, PA2, PA3 and PA4 which refer to masquerading, man-in-the-middle, replay and node tampering attacks respectively. As depicted in the table II, our proposed protocol fares well in all these security features and is secure against all the mentioned attacks. Most other mentioned protocols provide mutual authentication, while [25] provides one-sided authentication only. For secure authentication and communication, it is desirable to generate a unique session key for each session. Only the protocols in [28–30] are capable of this. Physical security is a very important security metric to be considered for IoT environments, where the devices like UAVs are deployed in hostile environments and are thus vulnerable to physical tampering and device capture. Other than our proposed protocol, the only other protocol which accounts for physical security of the individual UAVs is [27]. In [27], upon device capture, the UAV hardware is disabled by sending a destruction signal from the ground station, to protect it from data theft. Further, most of these works only perform some type of informal security analysis on their protocols. For any security protocol, a thorough debugging analysis is crucial to ensure that the protocol is not susceptible to exploitation by malicious entities and also to ensure that vital information is not leaked or altered. In this paper, a Mao Boyd logic [24] based formal protocol proof is shown to verify the proposed protocol. The protocols which provide an informal security analysis are shown with a * mark, while those without any discussion on the security analysis are marked with a cross. Most of the mentioned protocols fail to provide protection against one or more important security attacks as shown in the table. Our protocol is not only safe against the common attacks like masquerading attack, MITM attack and the replay attack, but also against node tampering attack. Even if an adversary physically captures a UAV from this network, any attempt to tamper with the PUF will render the PUF unusable and therefore the captured UAV will not be able to authenticate with the base station. Thus, the adversary cannot resort to nefarious activities even in the event of a device capture.

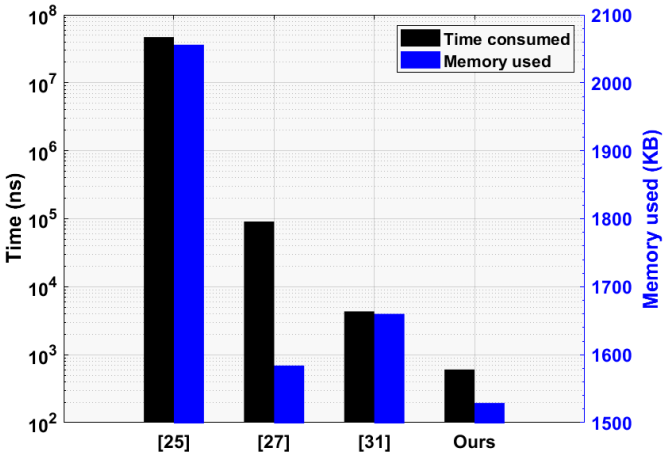


Fig. 6: Performance comparison with popular encryption schemes

B. Performance Comparison

In Fig. 6, we have compared the time consumed and memory used for the encryption process in our protocol with the schemes in [25], [27] and [31]. The protocol was implemented in C and compiled using GCC v7.4.0 on the Ubuntu 18.04 operating system. The simulation environment used was a 64-bit Intel(R) Core(TM) i7-5500 CPU with 8GB RAM. While [25], [27] and [31] took 4.7×10^7 ns, 90890 ns and 4340 ns, our protocol took just 605 ns to execute. Thus, our protocol performs exponentially better than the above mentioned schemes in terms of execution time. While [25], [27] and [31] consumed 2056 KB, 1584 KB and 1660 KB of memory respectively, our protocol required only 1429 KB showing that our protocol utilizes memory more efficiently. Therefore, our scheme is a better choice considering both the processing capabilities and the memory constraint of UAVs.

VII. Conclusions

This paper proposed DRONA, a PUF based mutual authentication protocol for UAV networks. DRONA uses a

challenge-response scheme enabled by PUFs which gives it the advantage of not having to store any secret information in the UAV nodes. Only one challenge-response pair per UAV is stored in the base station. After a UAV authenticates with the base station, a secure session key is established between them. We showed that DRONA has mutual authentication, untraceability and physical security along with protection against various attacks such as masquerade, MITM, replay and node tampering attacks. It is also capable of generating a unique session key for each new session. Moreover, it outperforms other existing solutions in terms of time consumption and memory usage. Hence, we argue that the proposed protocol, DRONA is very efficient and viable for UAV networks.

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